From Data Deluge to Data Strategy: Making Sense of Music Numbers in the Age of Platforms and AI

Part 1: The Data Deluge. A literature-grounded expansion of "From Data Deluge to Data Strategy: Get the Power of Insights," presented by Isabella Pighi and Chiara Santoro at SXSW 2025 (Music and Tech track), updated to mid-2026.

Abstract

Every time you play a song, skip it three seconds in, save it, share it, or follow the artist, that action gets logged, bundled with millions of others, and sold back to the industry as market intelligence. Music is now one of the most heavily measured things people do for fun. The argument of this piece, first made on stage in 2024, is that all this data is not automatically an asset. When the numbers are incomplete, skewed, or read the wrong way, they produce decisions that are confident and wrong. And the same platforms that hand the industry its data are also quietly shaping the behaviour those numbers record. This paper puts that argument on firmer footing, drawing on research from media studies, recommender-systems engineering, and fan studies, and it brings the story up to mid-2026. A lot changed in a short span of years. A major label pulled its whole catalogue off the biggest discovery app and let everyone watch what happened. The definition of a paid stream was rewritten. The "superfan" went from a line on a slide to the centre of the industry's growth plans. And AI-made music started pouring into streaming libraries, some of it collecting royalties through outright fraud. The lesson has not changed, but the stakes have risen: the value of music data comes not from how much of it you have, but from the care you take in collecting it, comparing it fairly, and deciding what you are allowed to do with it. That is the difference between a deluge, which is just signals piling up because the platforms emit them, and a strategy, which is knowing what each signal means and who it serves before you hand it to the machines that increasingly make the calls.

---

1. More data than ever, and less certainty

Picture the last song that blew up seemingly out of nowhere. Somewhere in a label office, an analyst is staring at a spike on a dashboard and trying to answer one question: is this real, and can we bet money on it? That question is harder than the flood of available numbers makes it look, and understanding why is the whole point of this paper.

Researchers have a word for the process of turning everyday life into countable data: *datafication*. van Dijck (2014) uses it to describe the belief that almost any human activity can be captured as data and traded, and she warns about the mindset that comes with it, a faith she calls "dataism" that treats those numbers as plain facts of nature rather than things somebody built and shaped. Hold on to that distinction, because everything else rests on it. The trouble with a song that has a million streams is not only that most of those plays might be half-second skips. It is that the number itself tempts you to believe "streams" is a natural truth about how people listen, when it is really a record of how one particular app decided to count one particular kind of attention. Nieborg and Poell (2018) push this further for creative work specifically. They argue that making and selling culture now depends on the platforms, their data, and their marketplaces, which means any list of "where music data comes from" is really a map of who the industry depends on.

The original talk boiled the danger down to three ways good data goes bad. The first is the oldest rule in computing, garbage in, garbage out: a decision is only as good as what you feed it, so a headline number stripped of context (who actually listened, for how long, whether they ever came back) is a liability wearing the costume of an asset. The second is mistaking things that happen together for things that cause each other. A streaming jump that lines up with a burst of TikTok posts invites a tidy story where the posts drove the streams, when the likelier engine was an editorial playlist add that nobody put in the slide deck. The third, and the one that does the most damage, is the self-fulfilling prophecy: recommendation systems that push already-popular songs precisely because they are already popular, which cements the winners in place and quietly strangles the discovery those systems are supposed to deliver. Each of these has since been pinned down with real evidence, and this paper walks through it.

Why revisit all this now? Because the facts on the ground moved. In 2024, TikTok was the undisputed engine of music discovery, a "stream" meant more or less the same thing everywhere, superfans were a hunch, and AI music was a novelty act. In just a few years, a major label yanked its entire catalogue off TikTok and researchers got to measure the fallout, Spotify started refusing to pay royalties on tracks below a play-count cutoff, the superfan became the hinge of the industry's growth forecasts, and one streaming service admitted that close to half its daily uploads were fully AI-generated. None of these are side notes. They are a stress test of the 2024 argument, and each one confirms where it was pointing while raising what is on the line. The rest of the paper takes the argument apart section by section, backing up what was claimed and updating what has shifted underfoot.

2. How we ended up measuring everything

The machinery behind today's flood was built piece by piece to solve a much smaller problem: making sure the right people got paid. Weekly popularity tracking is older than it looks: *Billboard* published its first chart, a ranking of best-selling sheet music, in July 1913, and its first chart based on record sales on 4 January 1936. The real turning point came in 1991, when *Billboard* began building its album chart from SoundScan, a system created not by the industry but by two outsiders, the industry veteran Michael Shalett and the pollster Michael Fine, which swapped guesswork and reported estimates for barcode scans at the register (The Ringer, May 2021). SoundScan was an independent company at the time, and Nielsen acquired it later, which is why the system is usually remembered as Nielsen SoundScan. Adoption was phased rather than instant, with the album chart switching in May 1991 and the Hot 100 following that November. Overnight the charts became a record of actual transactions instead of the industry's own say-so. The internet added a second reason to keep score, tracking not just what people bought but what they listened to, and the signals multiplied fast. BigChampagne started reporting the most-shared files on Napster. Then came iTunes in 2001, MySpace in 2003, YouTube in 2005, Twitter and Spotify in 2006, and eventually TikTok, each one turning music into something you navigate through enormous libraries, where the act of navigating throws off data at every step. If you want people to find their way through a hundred million tracks, you have to give them playlists, recommendations, and search, and every one of those tools is also a sensor quietly writing down what they do. That is how the industry ended up with the kind of behavioural analytics that tech companies live on, not just sales figures.

This measuring is not a harmless byproduct. It is a form of power, and that is the case Drott (2018) makes in describing streaming as a kind of surveillance, then develops fully in his book *Streaming Music, Streaming Capital* (2024): the listening data is not exhaust from the engine, it is the thing being sold. If you want to see how tangled and secretive one platform's counting machine really is, the team behind *Spotify Teardown* reverse-engineered it from the outside (Eriksson and colleagues, 2019), and their work is the reason it is fair to treat platform metrics as black boxes rather than clear windows onto reality.

The mergers the talk listed tell the same story of an industry growing up around this data. Live Nation swallowed BigChampagne in 2011, Pandora bought Next Big Sound in 2015, Apple picked up Semetric (the company behind Musicmetric), and Nielsen's music arm was sold to *Billboard*'s parent company in 2019, becoming MRC Data before being renamed Luminate in 2022 (Billboard). The measurement business thereby ended up under the same roof as the chart it feeds. Meanwhile a set of independent analytics firms, Chartmetric, Soundcharts, and Songstats among them, compete with the old data houses to turn a scattered mess of signals into something worth paying for. One line from the talk deserves a note of caution. The often-repeated claim that music is now worth more than film traces to economist Will Page's annual *Global Value of Music Copyright* report, which put the total value of music copyright at $45.5bn for 2023, ahead of the $33.2bn global cinema box office (Music Ally, Nov 2024). It is best read with care rather than as a settled fact: the $45.5bn combines recorded music, collective-management-organisation collections, and publisher income, and recorded music alone (about $28.5bn) actually sits below the box office, so the topline is neither a recorded-music figure nor a strict like-for-like comparison. The overall direction, though, is not in question. Measuring music has gone from a quiet back-office job about royalties to a front-office business about intelligence.

3. Where the numbers come from, and why that matters

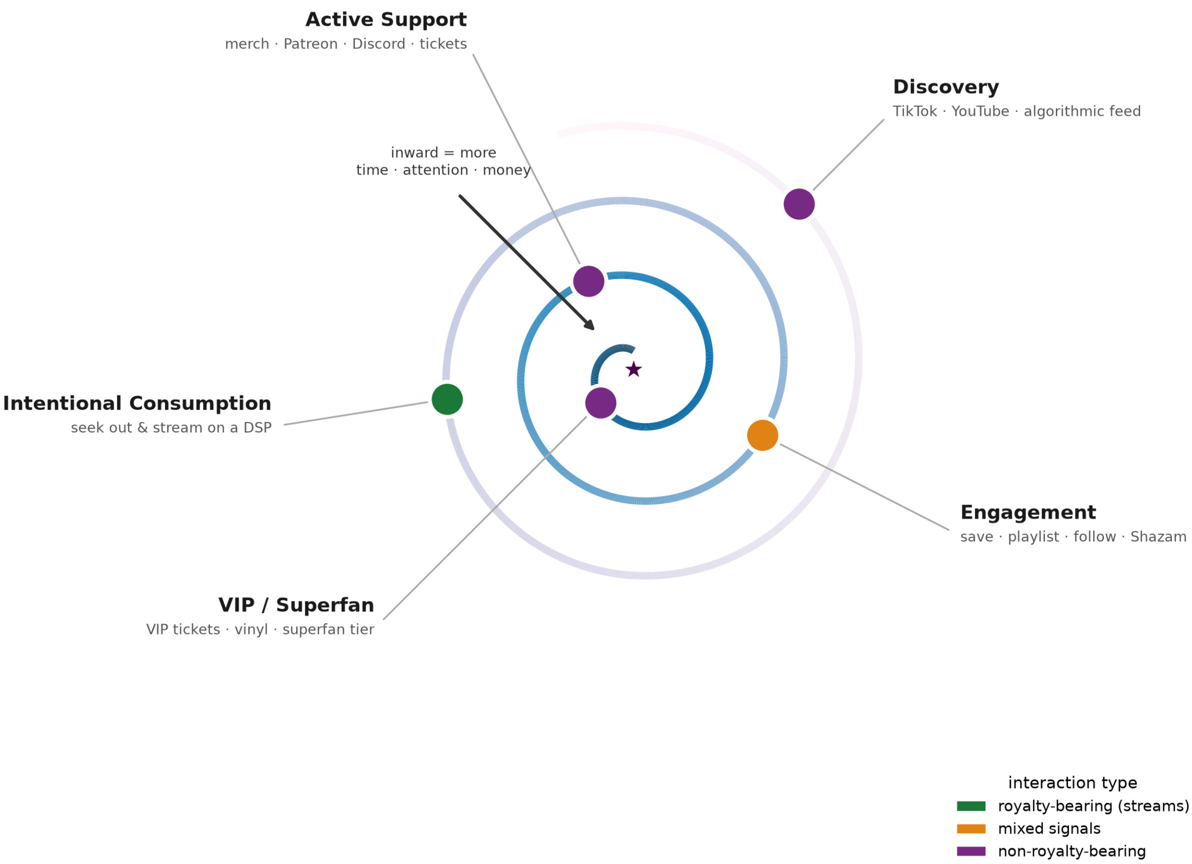
The talk sorted music data into four buckets: the streaming services themselves, social media, the charts, and everything else that signals popularity without being a play (press coverage, Wikipedia traffic, concert tickets, merch sales, Discord servers, Patreon memberships). As an inventory it holds up fine. What the research adds is a warning: these are not four independent taps drawing from one shared pool of truth. They are outputs of platforms whose own interests decide what gets measured and how. Morris (2015) calls the companies that mine listening data to package up "taste" *infomediaries*, and shows that what they do is not neutral reporting but a kind of manufacturing, taste made by code. Prey (2017) sharpens the point into what he calls *algorithmic individuation*: streaming apps do not simply serve an audience that already exists out there, they use data to conjure individual listener-profiles into being. Which quietly knocks the legs out from under the comfortable assumption that a "fan" is a fixed thing you can just tally up.

The charts deserve their own health warning, precisely because people trust them most. A chart is not a measurement. It is a stack of other measurements piled on top of each other, which means it soaks up every bias and every bit of gaming in its ingredients and then serves the result as one clean, authoritative number. Today's gatekeeping is a blend of human editors and algorithms, and Bonini and Gandini (2019) capture it with a memorable finding: the first week of a song's life on a platform tends to be editorial, the second week algorithmic. That is the correlation-versus-causation trap in action, because a "viral" spike may just be the mechanical result of an editorial playlist placement handing the baton to an algorithmic one. Prey and colleagues (2020) show where that curatorial power actually sits, inside the playlists, and Maasø and Spilker (2022) give the tangle of human and machine gatekeeping a name, the "streaming paradox." The takeaway for anyone building decisions on this data is blunt: having many sources is not the same as having independent ones. If you average together ten platforms that all lean on the same popularity signals, you do not cancel out the bias, you just spread it around and make it look objective.

4. Why fans don't fit in a funnel

Marketing people love a funnel: reach at the top, then consumption, then engagement, then fandom at the bottom, with money spent as the finish line. It works for selling toothpaste because a shopper's relationship to toothpaste is mostly a transaction that moves in one direction. Music is not toothpaste. A fan is not a customer, and the reason goes back to Jenkins (2006), who showed that fans are co-creators rather than end-of-the-line consumers, and that their most valuable behaviour is often making things (edits, covers, memes, fan art) rather than buying things. Force that relationship into a funnel and you get its shape wrong. What fans actually do is loop: they discover an artist, dig in, drift away, come back, and fall deeper. It looks less like a funnel and more like a *spiral* that a listener travels around again and again, giving up a little more time, attention, and money on each pass (Figure 1).

The fandom spiral: engagement deepens inward, but its signals are not commensurable



The highest-value stages — Active Support and VIP/Superfan — are largely non-royalty-bearing. This is the core measurement problem.

The spiral makes visible a measurement problem the funnel hides, which is that its stages cannot be compared on the same scale. A stream on a streaming service pays a royalty and gets logged to the decimal. A late-night listening session on someone's Discord, a reply to an artist's post, a night out at a show: none of that pays a per-play royalty, and much of it is not logged anywhere at all. Yet to the artist, that unlogged moment might be worth more than a thousand streams. The research that makes the case for taking these off-the-books interactions seriously is Nancy Baym's work on what she calls *relational labour*. In Baym (2015) and her book *Playing to the Crowd* (2018), she shows that the ongoing, personal connection work musicians do with their audiences online is genuine labour and is the very substance of modern fandom, which is exactly why a metric that only counts paid plays is measuring the wrong thing. The emotional pull that drives fans to show up is theorised by Papacharissi (2014), and the one-sided intimacy that turns a casual listener into a devoted superfan is the parasocial bond studied by Chung and Cho (2017).

So how do you compare the worth of a stream to the worth of a fan's emotional investment or their ticket to a sold-out room? The talk raised the question and could not answer it, and honestly, neither can anyone else with a single tidy formula. But the research offers a disciplined way to think about it. Trunfio and Rossi (2021) reviewed the field and found that "engagement" gets measured inconsistently even in the studies that depend on it, and Scharlach and Hallinan (2023) supply the key idea in their study of what engagement buttons actually mean: a like, a save, and a share are understood by people to signify different things, so mashing them into one "engagement" score is a mistake. The sensible move is not to invent a universal currency but to refuse to. Name what each signal is for, keep paid interactions and unpaid ones in separate columns, and only compare things that are genuinely alike. The fact that the most valuable part of the spiral sits squarely in the unpaid column is not a hole to be plugged with a cleverer metric. It is a permanent feature of how music works, and one the industry is only now, belatedly, trying to instrument (Section 7).

5. The self-fulfilling prophecy, spelled out

Of the talk's three failure modes, the third one, algorithms that boost the already-popular and lock it in place, was the boldest claim and the one with the least proof at the time. The research has since made it precise, and precision makes it stronger, not weaker. The cornerstone is Chaney, Stewart and Engelhardt (2018), who named the problem *algorithmic confounding*. When a recommender learns from data that its own past recommendations created, the loop feeds on itself: everyone's taste gets more alike, and, strikingly, the system actually gets worse at the job it was built to do. The loop is not just unfair to small artists, it is self-defeating for the platform. Mansoury and colleagues (2020) showed the mechanism in action, with recommendation feedback loops amplifying popularity bias and steadily shrinking the range of things people ever get shown. If you want the full map of the problem, the biases at work and the fixes proposed for them, the survey by Chen and colleagues (2023) is the place to start, and it is what turns a vague worry into a well-understood engineering challenge.

The specific shape the loop reinforces, a handful of superstars and an enormous tail of everyone else, was measured in music before it was theorised in general. Celma and Cano (2008) traced how popularity concentrates in the recommendation network, and Celma's book *Music Recommendation and Discovery* (2010) is still the standard treatment of the long-tail problem behind the talk's worry about emerging artists. But the evidence does not all point one way, and grounding the argument honestly means saying so. Datta, Knox and Bronnenberg (2018) found that moving to streaming actually *widened* what individual listeners explored compared with owning music, and the big Spotify study by Anderson and colleagues (2020), the empirical heart of this section, compared algorithm-fed listening against self-chosen listening and found that the algorithm boosted how much people consumed but was linked to *less* variety in what they heard. So the effect the talk warned about is real and measurable, but it is a tension rather than a verdict, because the same systems that homogenise can also expose someone to more than they would ever have found on their own. Villermet and colleagues (2021) add a necessary caution by carefully separating human choices from algorithmic ones in on-platform listening, and finding that "the algorithm did it" is often an attribution error, which is the correlation-versus-causation warning aimed back at the algorithm itself.

Where the field has moved furthest since the talk is fairness, and this is the best place to update the argument. The research agenda was set by Schedl and colleagues (2018), and popularity bias got a concrete measuring stick, a country-by-country "mainstreaminess" score, from Bauer and Schedl (2019). The work on gender is directly on point for the talk's worry about skewed samples: Ferraro, Serra and Bauer (2021) document a feedback loop that entrenches gender imbalance in music recommenders and argue for stepping in to break it, and Melchiorre and colleagues (2021) test the gender fairness of these algorithms head-on. The wider menu of fairness methods is mapped by Deldjoo and colleagues (2023), and the closest thing to an answer for the talk's plea to "minimise the evergreen effect" is the long-run study by Porcaro, Gómez and Castillo (2024) on how recommendation variety actually shapes listeners over time. One small correction tightens the argument. The talk used "filter bubble" and "echo chamber" loosely, but Nguyen (2018) draws a useful line between an *echo chamber*, which actively runs down outside voices, and an *epistemic bubble*, which simply leaves them out. Music recommendation builds the second kind, and that matters, because a bubble can be popped by exposure while an echo chamber cannot.

6. Four kinds of trouble

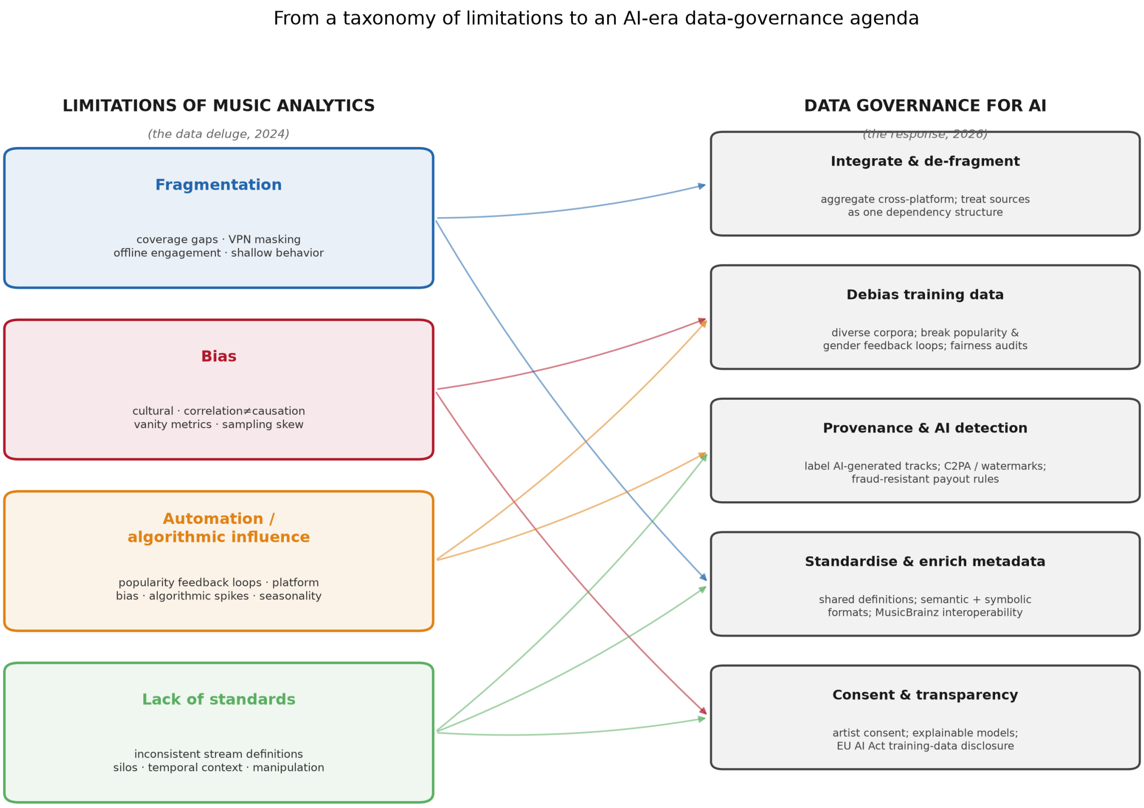
The obstacles to reliable music analytics come in four families, and naming them precisely is what turns a vague unease about "data quality" into an actual to-do list (Figure 2, left).

Fragmentation is the scattered, patchy state of the signal itself. Not every platform reports every artist, VPNs scramble the geography, offline listening is mostly invisible, and even where behaviour is logged the logs are thin: a skip gets recorded, but the reason for it does not.

Bias works on interpretation. Metrics miss cultural and regional nuance, correlation gets read as causation, vanity numbers (followers, likes, playlist adds) stand in for engagement they do not actually measure, and the crowd generating the data is not the crowd you care about. TikTok's user base, for instance, skews young and female in ways that do not represent the listeners for plenty of genres.

Automation and algorithmic influence is the family Section 5 lays out in detail: systems that cannot tell passive listening from active, that favour the mainstream, that produce spikes and dips having nothing to do with how good an artist is, and that get muddied further by seasonality.

Lack of standards is the missing shared vocabulary. A "stream" is not the same event on Spotify as on Apple Music, data sits siloed by platform, short measurement windows hide a song's real lifecycle, and every number is open to manipulation through playlist gaming, bots, and coordinated campaigns.

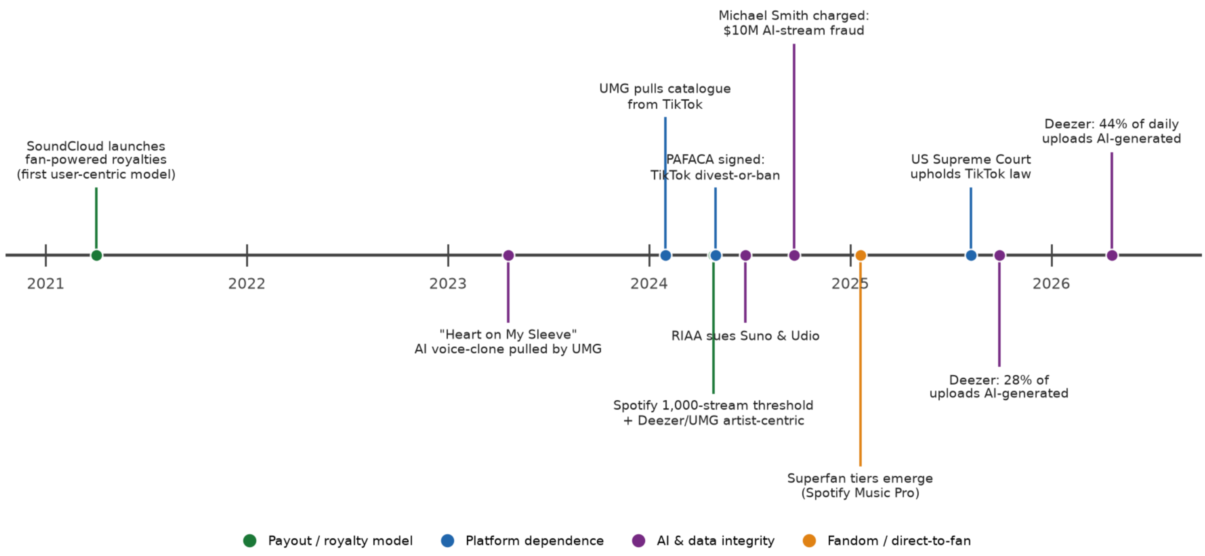


These four are not really separate, and the most useful way to hold them is as a checklist of demands on any attempt to compare numbers fairly. The talk's own diagnosis was that every normalisation effort so far has been local and arbitrary, with proprietary indices from Chartmetric, Luminate, and others boiling a many-sided performance down to one debatable score, and the research backs that scepticism up. A real normalisation would need shared definitions across platforms, honest blending of streaming with social, live, and offline signals, room for qualitative and contextual data alongside the counts, and transparency into the algorithms and revenue splits that produce the numbers in the first place. The recommended course corrections fall out of that directly: reward the quality of engagement over raw volume; play down absolute popularity in favour of growth and potential; fold in demographic and regional context; actively defuse algorithmic bias with diverse training data and audits; separate a catalogue's evergreen performance from a new release's momentum so old totals do not drown out fresh signal; and treat online impact and offline impact as two distinct measurements rather than stand-ins for each other. None of these is a formula. Each is a discipline, a decision about what to measure and what to refuse to blur together, which is exactly what moving from deluge to strategy means.

7. What changed: the argument stress-tested, 2024 to 2026

The years bracketing the talk were not a quiet stretch where the argument sat undisturbed. A run of developments tested it directly, and each turned an abstract worry into a concrete, dated event. Laid out on a timeline (Figure 3), they cluster into four threads: how the money is split, how dependent the industry is on a handful of platforms, how fandom is being monetised, and how AI is reshaping both what gets made and what can be trusted.

A stress test in real time: the music-data era, 2021 to 2026



7.1 Platform dependence, finally measurable: UMG and TikTok

The clearest real-world demonstration of the talk's platform-dependence theme was the falling-out between Universal Music Group and TikTok in 2024. When their licence lapsed at the end of January, UMG refused to renew and pulled its recorded and publishing catalogue off the app, publishing an open letter that accused TikTok of underpaying, of tolerating AI-generated music that waters down the royalty pool, and of building a business on music it barely paid for (Variety, Jan 2024; Digital Music News, Jan 2024). For roughly three months, the catalogue of the biggest music company on earth went silent on the single most important place people discover songs. It was an accidental natural experiment in exactly the causal question the talk kept flagging. A Harvard Business School analysis of that silent window found effects that were real but uneven and hard to pin down (HBS Working Knowledge), which is itself the lesson: even a clean removal does not cleanly reveal what TikTok contributes, because too many other things move at the same time. The two companies struck a new deal in early May 2024, reportedly covering compensation and AI protections (Variety, May 2024).

The dependence runs deeper than any one licence, because the platform itself is now a political variable. The Protecting Americans from Foreign Adversary Controlled Applications Act was signed into law on 24 April 2024, requiring ByteDance to sell off TikTok's US operations or face a ban (PAFACA overview). TikTok's free-speech challenge failed when the Supreme Court unanimously upheld the law in *TikTok Inc. v. Garland* in January 2025 (Holland & Knight, Jan 2025; CRS, LSB11261). Enforcement then got pushed back repeatedly through 2025, ending in a September 2025 White House framework for a US-owned joint venture (White House, Sept 2025; Federal Register, 30 Sept 2025). For anyone planning a data strategy, the moral is stark. A primary channel for both discovering and measuring music can be legislated out from under the industry overnight, and that is a fragmentation and governance risk the original list of limitations never imagined.

7.2 The definition of a stream got rewritten

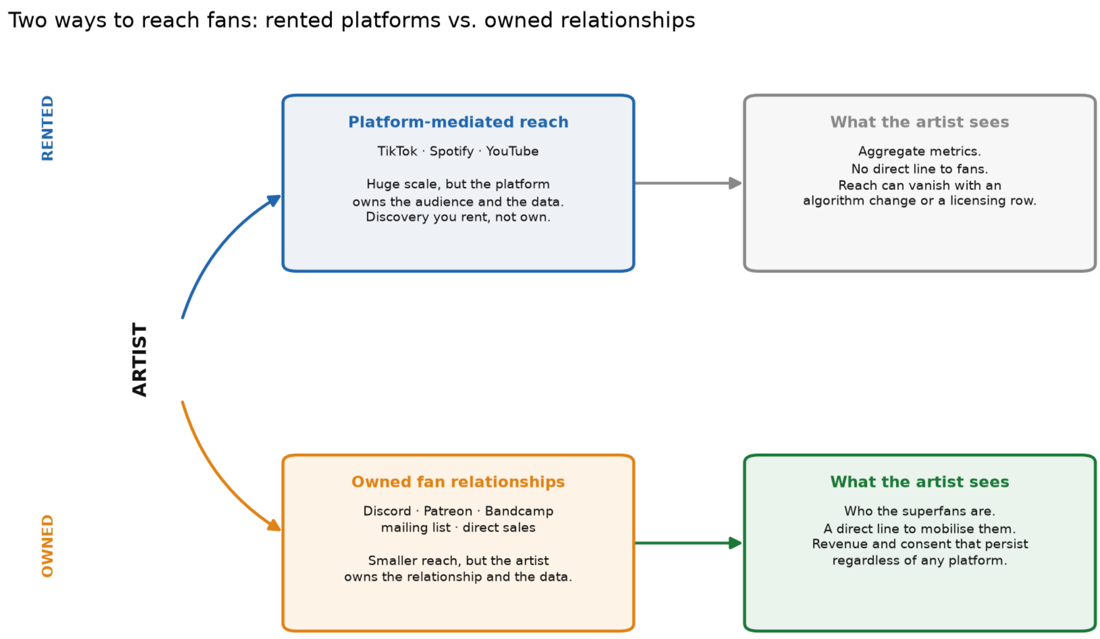
The talk's "lack of standards" worry picked up two concrete examples that change what a stream count even means. First, from early 2024, Spotify brought in a **1,000-stream annual threshold**: a track has to rack up at least a thousand plays in the previous twelve months before it earns any recorded royalty at all, part of a package that also charges distributors for detected fake streams and sets a minimum play length for functional audio like white noise (Music Business Worldwide; iMusician, Apr 2024; Spotify for Artists). A headline stream count no longer maps neatly onto money, so the talk's warning about "a million streams that are mostly skips" is now literally written into the payout rules. Second, the industry started rewriting the rule that turns a play into a payment. Today's headline example is the **Deezer and Universal "artist-centric" model**, announced in late 2023 and rolled out through 2024, which tilts the royalty pool toward professional artists and stops paying out on non-artist noise content, a direct attempt to defuse the passive-listening dilution the talk flagged (Music Week, 2023). But it is worth setting the record straight on where this idea came from, because it did not start with a major label. **SoundCloud got there first.** In April 2021 it became the first sizeable streaming service to switch a slice of its payouts to what it called **"fan-powered royalties,"** paying an artist out of what their own listeners spent rather than dividing one giant pool by total play count (Music Ally, Mar 2021; Rolling Stone, 2021). The distinction matters because it is the closest thing the streaming economy has to the fandom spiral made literal: your money follows the artists you actually listen to, instead of subsidising whatever the crowd is streaming most. By 2022 SoundCloud reported that around 135,000 artists were being paid this way (MBW, 2022). The Deezer and Universal move two years later took the same instinct and applied it at major-label scale. Both changes landed alongside subscription price rises and bundling, which make comparing per-stream value across platforms or across time harder still. The standards problem is getting worse, not better.

7.3 The superfan went from slide to strategy

The single biggest vindication of the fandom spiral is that its innermost stages, Active Support and VIP, have become the industry's central growth plan under the label of the *superfan*. Luminate's research popularised the finding that roughly **15% of US music listeners are superfans** who spend much more than average on music and artist products (MBW, 2023). Goldman Sachs's *Music in the Air* reports put superfan monetisation, emerging markets, and price rises at the centre of a global forecast reaching toward the $200bn range (Music Week, 2024; Forbes, Sept 2025). Labels put the idea into practice, and from February 2025 Spotify was reported to be preparing a pricier **"Music Pro"** superfan tier bundling early-access tickets, high-fidelity audio, and AI remix tools (Music Ally, Feb 2025; Fortune, Feb 2025). The thesis is contested, and by mid-2026 some trade commentators were arguing the standalone superfan subscription had over-promised (CelebrityAccess, May 2026). But that debate is downstream of the talk's exact point. The most valuable interactions are largely unpaid ones, and the industry is now racing to instrument and monetise the deep end of the spiral without having solved how to measure it.

7.4 Where superfans actually live: Discord, Patreon, and owned channels

If the superfan is the prize, the obvious next question is where you find one, and the answer is almost never on a streaming service. The deepest fandom has migrated to channels the artist can actually own, and this is where the earlier point about "popularity beyond listening" stops being a footnote and becomes the main event. Two platforms deserve far more attention than the talk gave them. **Patreon** lets fans pay a recurring subscription straight to an artist in exchange for exclusives and access, which turns the fuzzy idea of "active support" into a monthly number the artist controls and can see. **Discord** is where the community itself lives, a private clubhouse of servers and channels where fans talk to each other and, sometimes, to the artist, generating exactly the relational connection Baym described, at a scale a mailing list never could. Neither shows up in a stream count. Both sit in the innermost, highest-value rings of the fandom spiral, and both hand the artist something a platform play never does: a direct, owned line to the people who care most (Figure 4).



The deepest, highest-value fandom lives in owned channels. Fred again..'s Discord coordinated 18 fan-led launch parties; SoundCloud's fan-powered model paid artists by their own listeners, not the pool.

The clearest demonstration is **Fred again..**, whose rise was built on precisely this owned-community model. His Discord server became the operational hub of the fandom rather than a marketing afterthought. For the launch of *Actual Life 3* in October 2022 he did not lean on a playlist pitch; he used Discord to coordinate eighteen fan-led listening parties across cities worldwide the day before the record came out, letting the community itself stage the release (Music Ally, Nov 2022; Levellr case study). The same community muscle powered the spontaneous, hard-to-measure live moments he became known for, from a mass bike-ride rave to Marble Arch to the last-minute pop-up shows announced with almost no notice, where being in the right server at the right moment was the ticket (Where Music's Going).

By late 2025 he had scaled that instinct into a whole release strategy. Rather than tie the campaign to a conventional album, he branded the entire global run under the name of the evolving project itself, **USB002**, and built it as a "ten shows, ten cities, ten weeks" format in which a new track dropped in tandem with each show and venues were revealed only days ahead. The run opened in Glasgow on 3 October 2025, travelled through Madrid, Brussels, Lyon, Dublin, Toronto, Chicago, and San Francisco, and closed in Mexico City on 12 December, before he immediately extended it into a dual residency, in New York and then at London's Alexandra Palace in early 2026 (Billboard, Dec 2025; NME, Dec 2025; Music Ally, May 2026). The London finale drew a run of surprise guests, from The Streets and Underworld to La Roux, Skream & Benga, JME and D Double E, and closed with a rare back-to-back set alongside Daft Punk's Thomas Bangalter, only his second live appearance in nearly twenty years, filmed and premiered afterward on YouTube (Resident Advisor, Mar 2026). This is the fandom spiral turned into an operating model: the release is the tour, the tour is announced through the community, and the whole thing is documented for that community afterward. None of that activity is a stream. All of it is the fandom spiral in motion, and almost all of it is invisible to the metrics the rest of this paper has been interrogating. That invisibility is the whole point: the industry's growth story now depends on interactions its measurement systems were never built to see, which is exactly the gap Part 2 sets out to close.

7.5 AI music went from curiosity to systemic risk

The talk closed by asking whether AI would make the data deluge better or worse, and answered "it could go either way." Both directions are now visibly in play, and "garbage in, garbage out" has taken on a literal, adversarial form. On the input side, the fight over what data music-generation models are trained on became the defining battle of the period. In June 2024 the RIAA and the major labels filed landmark copyright suits against the two leading generators, **Suno** and **Udio**, alleging mass unlicensed copying of sound recordings for training (RIAA, Jun 2024). The defendants argued fair use while effectively conceding large-scale ingestion, and through late 2025 the disputes began to resolve into licensing deals rather than verdicts. Universal Music Group settled its case against Udio in October 2025, announcing a jointly built licensed AI music service for 2026 (Music Ally, Oct 2025); Warner Music Group settled with both Udio (mid-November 2025) and Suno (25 November 2025), striking licensing partnerships in each case (TechCrunch, Nov 2025; Music Business Worldwide, Nov 2025). Sony Music, by contrast, declined to settle and, as the last major label still in court, remained in active litigation against both generators, betting that a precedent-setting ruling would be worth more than any single settlement (Tech Times, Jun 2026). A key summary-judgment hearing in the Massachusetts Suno case was scheduled for July 2026, with a precedent-setting ruling on the core fair-use question still pending. The outcome will decide whether commercial music AI has to license and document what it trains on, which is to say whether provenance and consent metadata become mandatory rather than nice-to-have.

The intellectual-property threat runs wider than training data, though, and the case that made it vivid was **"Heart on My Sleeve."** In April 2023 an anonymous creator called Ghostwriter977 posted a track using AI to convincingly clone the voices of Drake and The Weeknd. It racked up millions of plays before Universal got it pulled across platforms (Variety, Apr 2023; NPR, Apr 2023). What made it a landmark was how awkward the takedown was to justify in law. A singing voice and a musical style are not themselves copyrightable, so the removal reportedly leaned on an incidental sample, a Metro Boomin producer tag, rather than on the impersonation itself, and the real legal recourse pointed toward publicity and likeness rights rather than copyright (Harvard Law Today, 2023). This is a different failure mode from an unlicensed training set. Here the harm is a synthetic performance that borrows an artist's identity, and the data problem underneath it is one of provenance and consent: nothing in the file said what it was or whose voice it used, so the metadata could not carry the very information rights depend on. It is the same integrity gap as the synthetic-uploads problem below, seen from the artist's side rather than the platform's.

On the output side, the scale of synthetic content is no longer a rounding error. Deezer became the first major service to publish hard numbers, reporting in September 2025 that about **28% of tracks delivered to it were fully AI-generated**, a share that had climbed to roughly **44% of daily uploads, on the order of 75,000 AI tracks a day, by April 2026** (Deezer Newsroom, Sept 2025; Deezer Newsroom, Apr 2026; TechCrunch, Apr 2026). Deezer answered with a detection and tagging system that labels fully AI-generated albums and keeps flagged content out of algorithmic and editorial recommendations (Deezer Community, Jun 2025). The emblematic case was "The Velvet Sundown," an AI-generated act that pulled large streaming numbers in mid-2025 before anyone confirmed it was synthetic (Euronews, Sept 2025). Listeners were consuming, and metrics were dutifully recording, music whose origin nobody had disclosed. That is the metadata-integrity failure the talk warned about, at industrial volume.

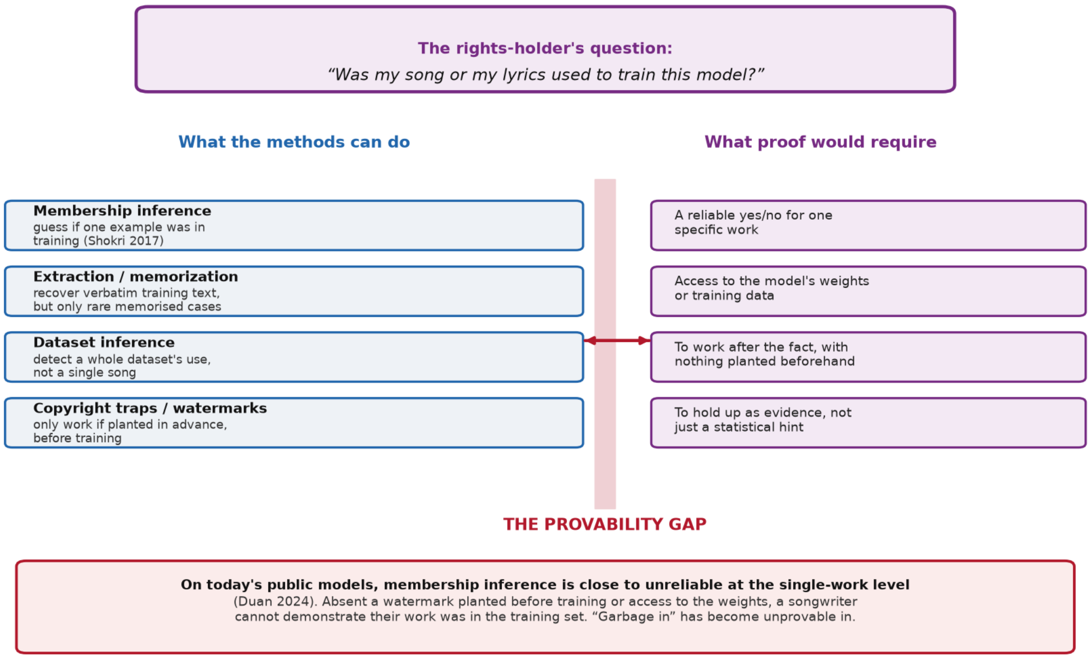
Where synthetic content meets the talk's manipulation family, you get a criminal case. In September 2024 the US Department of Justice indicted **Michael Smith** in what it called the first US criminal prosecution of its kind. He allegedly used bots to stream hundreds of thousands of AI-generated songs billions of times, fraudulently collecting roughly **$10 million in royalties** (DOJ SDNY, Sept 2024); he pleaded guilty in March 2026 (DOJ SDNY, Mar 2026; Rolling Stone). It is the talk's argument in its most concentrated form: AI generation on the supply side and bot consumption on the demand side team up to poison the very dataset that royalty payments and chart rankings run on. When both the content and the listening can be faked at scale, raw counts are not just noisy, they are actively corruptible by anyone with a reason to corrupt them.

The Smith case is unusual only in being prosecuted; the economics behind it are systemic, and they are driven by how streaming pays. Most major services use a pooled, or "streamshare," model: subscription and advertising revenue is collected into a single pot and distributed to rights-holders in proportion to their share of total streams (Rolling Stone, 2024). A payout is therefore not a function of what an artist's own listeners paid, but of that artist's fraction of everyone's plays, which turns each fake stream into a transfer from the shared pool rather than a victimless invention of new money. The scale is contested and hard to measure, which is itself telling: the fraud-detection firm Beatdapp estimates that at least 10% of all streams are fraudulent, implying annual losses of roughly $2 billion to $3 billion, while a more conservative 2023 study in France found only 1% to 3% of streams there were fake, which scaled globally would still divert up to around $510 million a year. Against the $14.9 billion that recorded-music streaming generated in the US in 2024, even the low estimate is a substantial sum redistributed away from artists with real audiences, and the cases are turning up worldwide, from a two-year prison sentence in Denmark to an alleged 28-million-play operation in Brazil (Rolling Stone, 2024). Cheap generative audio has driven the marginal cost of the fake "product" toward zero, so the incentive the pooled model creates is now trivially easy to act on.

7.6 The question you cannot answer: was my music used to train this?

Underneath the lawsuits and the takedowns sits a technical problem that ought to worry anyone who cares about where music data goes, and it is the one least understood outside computer-science circles. Suppose you are a songwriter and you suspect a generative model was trained on your catalogue. How would you prove it? The uncomfortable answer, on the models being sued today, is that you almost certainly cannot. This is not a gap in the law. It is a gap in what is technically possible, and it turns the talk's oldest principle inside out: "garbage in, garbage out" assumed you could at least see what went in, but with these models you often cannot see the inputs at all (Figure 5).

Why you cannot prove a song trained a model



The research area that tries to answer "was this specific example in the training set" is called **membership inference**, and it goes back to Shokri and colleagues (2017), who showed it could be done against machine-learning models under favourable conditions. The trouble is that those conditions do not hold for the huge models trained once on an enormous, near-web-scale corpus, which is exactly the situation for commercial music and language AI. Duan and colleagues (2024) tested membership inference on large language models directly and found it barely beats a coin toss, because a model trained for a single pass over a vast dataset simply does not memorise any one example strongly enough to leave a reliable fingerprint. A related line of work shows models *can* be made to regurgitate training data verbatim, from Carlini and colleagues (2021) on language models to Carlini and colleagues (2023) on image diffusion models, but Carlini and colleagues (2022) established that this memorisation is the exception, concentrated on the handful of examples that appeared many times, not the typical song buried once in the training set.

The workarounds only sharpen the point. Maini and colleagues (2021) proposed **dataset inference**, which can detect whether a whole dataset was used with far more confidence than any single record, and extended it to language models in Maini and colleagues (2024); useful for a label with a large catalogue, useless for one songwriter. Shi and colleagues (2023) built a detector for whether text was in a model's pre-training data, but it too is a statistical signal rather than proof. And the most reliable method needs foresight: Meeus and colleagues (2024) show you can plant "copyright traps," unique markers seeded into your work beforehand, so their later appearance betrays training use, but that only protects material you booby-trapped in advance, which is no help for a back catalogue already on the open web. Two 2024 results turn this from a catalogue of practical difficulties into a structural argument. Meeus and colleagues (2024), in a systematisation pointedly titled *Membership Inference Attacks on LLMs are Rushing Nowhere*, show that most recent membership-inference methods are evaluated on members and non-members assembled after a model's release, which introduces a distribution shift that inflates apparent success: the attack often detects when or what a text is, not whether it was trained on. And Zhang, Das, Kamath and Tramèr (2024) make the deeper point in a paper whose title states the conclusion, *Membership Inference Attacks Cannot Prove that a Model Was Trained On Your Data*: a credible training-data proof requires demonstrating a low false-positive rate under the null hypothesis that the model was *not* trained on the target data, but sampling that null is infeasible because it would require knowing the exact training set or retraining a foundation model from scratch. The proof a rights-holder needs cannot, even in principle, be built by inspecting the model from outside. Put together, the picture is stark. Unless you have access to the model's weights, or you watermarked your work before it was ever scraped, you cannot demonstrate after the fact that a specific song or lyric trained a model. That is why the governance response in the next section leans so heavily on provenance and consent recorded at the source. If you cannot prove misuse by inspecting the model, the only durable defence is to attach machine-readable rights to the music before it ever reaches one.

8. Governing the data an AI will run on

If decisions are increasingly handed to machines, then the quality, diversity, and origin of the data you feed those machines stops being housekeeping and becomes the whole game. The governance agenda maps straight onto the four families of trouble (Figure 2, right), and the events of Sections 7.1 through 7.4 have bumped several items from good practice to emerging obligation. Against **fragmentation**, the answer is genuine cross-platform integration that treats sources as one web of dependencies rather than independent truths. Against **bias**, it is cleaning up the training data itself, with diverse datasets, deliberate interruption of the popularity and gender feedback loops from Section 5, and regular fairness audits. Against the opacity of **automation**, it is provenance and detection infrastructure. Against the **lack of standards**, it is shared definitions and richer, interoperable metadata. And running through all four is consent and transparency: artists consenting to how their work trains models, models people can actually inspect rather than black boxes, and disclosure of what went into the training set.

That last item is now partly written into law. The EU AI Act took force on 1 August 2024, and its rules for general-purpose AI models, including a duty to publish a detailed enough summary of training data and to adopt a policy for respecting EU copyright and its text-and-data-mining opt-outs, began applying from 2 August 2025, run through a code of practice and a mandatory public disclosure template (European Commission, AI Act; European Commission, GPAI Code of Practice; Clifford Chance, Oct 2025). For music, this is the first regime that could actually force a model to disclose whether copyrighted recordings sit in its training set, the governance mechanism the talk called for, now starting to carry legal weight rather than just moral weight.

The technical toolkit for all this is real and already in use, which is what separates it from a wish list. The emerging data formats the talk gestured at are now mature research areas. Splitting a track into stems, the raw material for remixing, remastering, and the "AI remix" features showing up in superfan tiers, is handled by fast pretrained tools like Spleeter and higher-quality models like HT-Demucs, with the state of the art tracked by the Sound Demixing Challenge 2023. Symbolic and MIDI formats that make music editable, structured data for a model are advanced by MusicBERT for understanding symbolic music at scale and the Compound Word Transformer for generating whole songs. Most important for the integrity mess of Section 7.4 is provenance and watermarking. The cross-industry C2PA / Content Credentials standard attaches cryptographically signed origin information to media, with adoption urged by US NSA/CISA guidance, while on the audio side AudioSeal hides detectable watermarks in AI audio, AudioMarkBench tests how robust those watermarks are, and newer work aims at watermarking the training data of music-generation models so that whatever they output stays traceable. Even the economics of the fraud in Section 7.3 has a formal treatment now, in work on fraud-proof revenue division on subscription platforms. Together with open metadata efforts like MusicBrainz, these are the practical tools for governing music data rather than just piling it up.

9. From deluge to strategy

The argument of the original talk survives its stress test, and the run of events around it has sharpened the case rather than softened it. Music data is abundant and getting more so, and that abundance is still a trap for anyone who mistakes volume for insight. What has changed is that every one of the talk's warnings now has a concrete, dated example attached. Platform dependence has a natural experiment (UMG and TikTok) and a legal cliff-edge (PAFACA). The ambiguity of a "stream" has been written into payout rules. The unpaid deep end of the fandom spiral has become the industry's growth story under the superfan banner. And the manipulation-and-metadata problem has scaled into a world where nearly half the new uploads on at least one platform are synthetic, and where faked listening has already produced a criminal conviction. The research, from van Dijck and Drott through Chaney and Anderson to Baym and Scharlach and Hallinan, turns what were once slides into claims you can defend: that metrics are built rather than found, that feedback loops are self-defeating and not just unfair, and that the signals a fan throws off cannot be added up on a single scale and should not be forced onto one.

The strategy the title promises is not a dashboard or a proprietary index. It is a stance toward data. Decide what each signal means before you act on it. Keep things that cannot be compared in separate columns. Insist on knowing where your training data came from and whether you had the right to use it. And prefer the discipline of asking the right question to the comfort of a big number. The industry now has both the pressure to change, from AI decision-making, from regulatory disclosure, from industrial-scale fraud, and the tools to do it, from fairness methods to source separation to symbolic formats to content provenance to watermarking. Whether the data deluge gets better or worse with AI still comes down, as the talk said, to what goes in. The difference is that the choice is no longer hypothetical, and the cost of getting it wrong is no longer a bad marketing call. It is a set of machines that will make those calls at scale, on everyone's behalf, from whatever we hand them.

A companion bibliography of all verified sources, organised by theme with resolvable DOIs and dated URLs, accompanies this paper. Part 2, "Data Strategy," lays out the operating playbook.